What Makes a Good Day: A Machine Learning Analysis of Longitudinal Behavioral Data

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Abstract. This study investigates whether a subjective measure of daily satisfaction can be predicted using longitudinal behavioral and physiological data from a quantified-self dataset. The dataset consists of 326 weekly observations collected from January 2020 to March 2026 from a personal "Data Journal" system which combines self-reported measures, wearable device metrics, and YouTube consumption. Twenty-eight features were utilized to predict a weekly average of nightly satisfaction scores on a scale from 1 to 10. Several machine learning approaches were utilized and evaluated, including linear regression, random forest regression, and recursive feature elimination. Model performance was evaluated against a naive model, which always predicted the average value using mean squared error (MSE). The best performing model, a random forest using recursive feature elimination achieved an MSE of 0.776, which was a 14.5% improvement over the baseline model. Feature importance analysis revealed that measures of activity, sleep, and YouTube consumption were the most predictive factors. However, the overall predictive capability was modest, suggesting that subjective well-being is influenced by factors not captured in the behavioral dataset. The results demonstrate both the potential and the limitations of combining quantified-self data with machine learning to predict good days and determine how to have them more often.

Keywords: quantified self · happiness · longitudinal · behavior study · subjective well-being

1 Introduction

One of the most universal goals of humanity is to live a happy and fulfilling life. Centuries ago philosophers treated happiness not only as a desirable outcome, but perhaps the central objective of human existence. Aristotle wrote in the *Nicomachean Ethics* "happiness is the meaning and purpose of life, the whole aim and end of human existence" [1]. The United States Declaration of Independence famously identifies "the pursuit of Happiness" as one of the fundamental rights [7]. These statements affirm the universality of the goal; but they do not provide guidance on the practical means by which individuals might achieve it.

Modern science has inquired into what makes for a satisfying life. The study of subjective well-being emerged as a formal field of research during the twentieth century, and has expanded in recent decades [2,3]. In psychology, subjective well-being is understood to be a combination of **life satisfaction**, positive emotional experiences, and the absence of negative affect [4].

Longitudinal studies have provided important insights into what may influence the well-being of individuals. The famous Harvard Study of Adult Development highlighted the importance of social relationships, physical health, and various life circumstances as having contributed to long-term happiness [11]. Surveys have been conducted on the topic, with answers ranging from income, personality traits, social connectedness, and perceived control over your life [6]. Data science and machine learning techniques have enabled new approaches for studying well-being. The data for these approaches have typically come from large survey datasets [9] or through the mining of language used in social media posts [5]. These studies have demonstrated the ability of computational methods to help uncover complex and hidden patterns between behavior and reported well-being.

Alongside these studies, the past decade has also seen the birth and rise of the **quantified-self movement**, wherein individuals intentionally collect data about their daily activities and physiological signals [10]. Proponents of these quantified-self practices aim to use systematic self-tracking as a mechanism to achieve better understanding of one's self, typically focusing on health or productivity. Applying quantified-self methodologies directly to happiness has been attempted at large scale, through the Track Your Happiness project, which purports to give you the data to produce valuable insights on your daily behaviors and life satisfaction [8].

If happiness is influenced by quantifiable behaviors, can those behaviors that lead to increased satisfaction be identified empirically using machine learning?

Being an advocate of the quantified-self movement, I designed and have maintained a robust structured personal data and journaling system for over a decade. My research question is simply this: can I use these **individual-level longitudinal data** to determine what behaviors are most strongly associated with reported satisfaction. The goal of this study is not meant to generalize across populations, but instead to determine if the approach can yield meaningful predictive relationships between satisfaction *(output)* and behaviors *(inputs)*.

The research question can be summarized as follows:

To what extent can machine learning models predict good, satisfying days using behavioral and physiological data collected through a quantified-self dataset?

2 Methodology

2.1 Dataset Description

The data used in this study originate from my personal **Data Journal**. The Data Journal began in 2013 with pen and paper, but has since evolved to a multifaceted digital system producing a number of relational tables covering various aspects of my life and behavior. Figure 1 below provides an overview of the system's main functional containers.

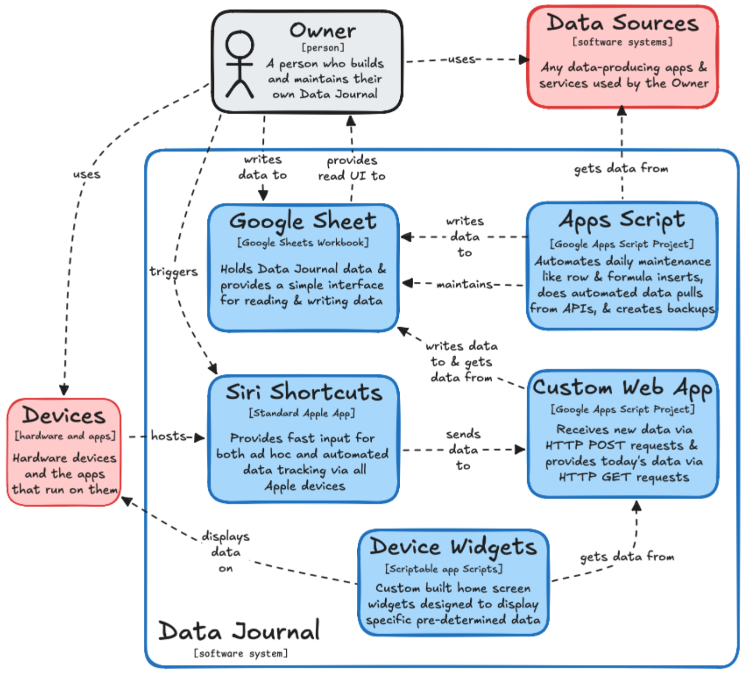

Fig. 1. Overview of the Data Journal system architecture, drawn in Excalidraw

While the Data Journal dates back to 2013 and includes data at levels of granularity from one datapoint per year down to many datapoints per day, the dataset used in this study is more reserved. Various measures were taken to protect my privacy and those of other individuals whose personal lives may have otherwise been inferred from a more robust representation of the underlying dataset. Section 2.2 covers data preprocessing in detail.

The dataset that was used for the purposes of this analysis covered **326 weeks of observations** from January 2020 to March 2026. Each observation included 28 behavioral and physiological variables, broadly including:

- Sleep metrics
- Wearable device scores
- Exercise counts
- Media consumption patterns
- Social and lifestyle behaviors
- Self-reported well-being measures

The dependent variable in this analysis is **satisfaction**, which is defined as a subjective nightly rating for how "good" the day was on a scale from 1 to 10, averaged over each week.

In detail, the independent variables in this analysis were:

Attribute	**Description**
summaries	count of daily summaries entered in a given week
at_home	count of days sleeping in my own bed
health	average of daily "how healthy do you feel" ratings
bedtime_decimal	average bedtime converted to decimal time
wake_decimal	average wake time converted to decimal time
sleep_duration	average hours slept
hr	average lowest heart rate
hrv	average heart rate variability
readiness	average Oura readiness score (0–100)
activity	average Oura activity score (0–100)
sleep	average Oura sleep score (0–100)
active_cals	average active calorie burn
pains	count of tracked instances of pain
workouts	count of workouts
strength	count of strength workouts
cardio	count of cardiovascular workouts
mobility	count of mobility workouts
ate_out	count of restaurant visits
outings	count of social outings
drinks	count of drinks consumed
quotes	count of captured quotes
events	count of tracked events
book	count of book reading sessions
tv	count of TV viewing sessions
movie	count of movies watched
videogame	count of videogame sessions
youtube	count of YouTube videos started

A detailed description of the Data Journal is available at `https://datajournal.guide`.

2.2 Data Preprocessing

I took several steps to sanitize the data and prepare it for this analysis. The initial phase of the data analysis involved two independent data preparation pipelines before joining the data and preparing it for machine learning.

The Data Journal pipeline picked up data from the "weeks" table, which aggregates data from the "days" table and tables covering more specific topics, such as "workouts". By using the aggregated data most of the sanitization was already complete.

Fig. 2. Data preprocessing pipeline, drawn in Excalidraw

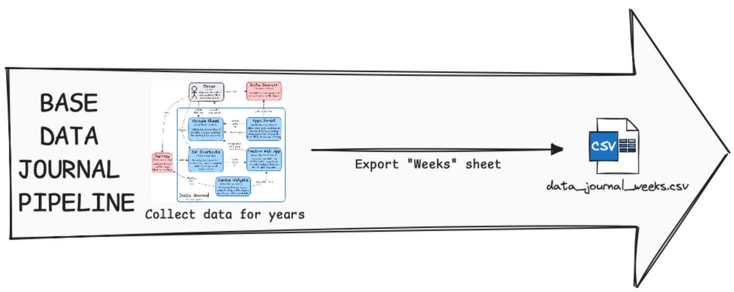

Fig. 3. Data Journal export pipeline, drawn in Excalidraw

Once the dataset was exported to `data_journal_week.csv`, a few redundant and otherwise uninteresting columns were removed to simplify the analysis.

While my Data Journal seeks to be a centralization of behavioral data, not every attribute in this project originated from self-tracking. Online services often track your usage of their platforms. In the European Union you are legally entitled access to these data. Some companies, such as Google, have taken the proactive step to make your data available to you upon request. I utilized **Google Takeout** to extract the **YouTube video views** it had tracked being made by my account over the time span covered in this project.

Detailed steps for this data extraction, including Python code are available at datajournal.guide. In short, using Google Takeout I exported my `My Activity` data feed. This produced a JSON file which I transformed to a CSV file using a Python script. This CSV contained the name and timestamp of every video that was watched by my YouTube account from January 2020 to March 2026. Using a PivotTable I aggregated the watch count to the week, and saved the file as `YouTube.csv` to be picked up by the next pipeline.

I then manually joined the two CSV files and further prepared them for analysis using the Jupyter Notebook available at:

`https://github.com/aarongilly/csis_capstone/blob/main/notebook.ipynb`

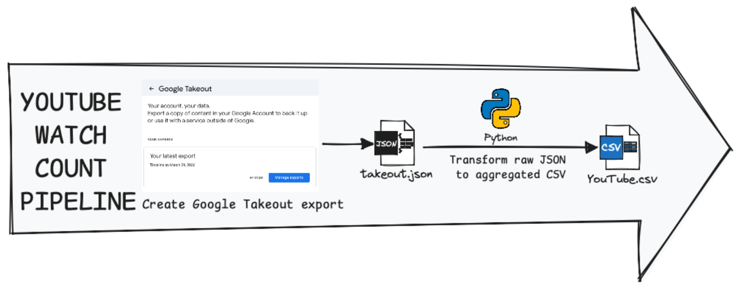

Fig. 4. Google Takeout export process, drawn in Excalidraw

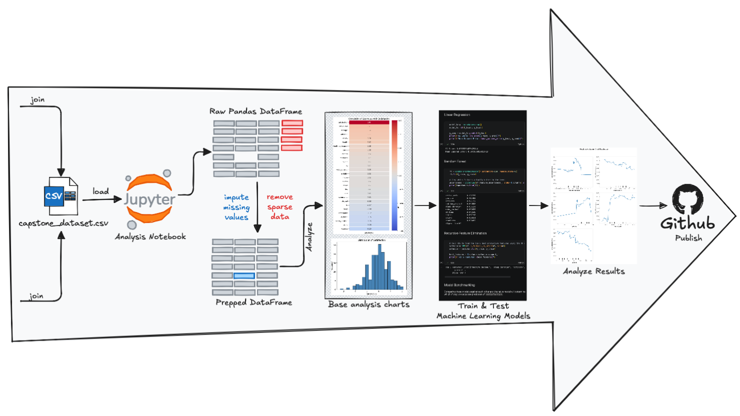

Fig. 5. Data joining and Jupyter Notebook process, drawn in Excalidraw

The non-informative identifiers were removed from the dataset. These variables represented the passage of time, but do not provide actionable predictive information. The identification of any temporal patterns could not be acted upon in the real world.

The dataset contained four columns which were deemed too sparse to be used for machine learning. All other columns had at most 2 missing values from the possible 326. All columns at this point were numeric, so the missing values were imputed as averages over the whole column. The impact of these 0.005% of imputed data is likely insubstantial.

After processing, the final dataset contained:

- **326 observations**
- **28 numeric features**

These features are described in Section 2.1.

2.3 Machine Learning Models

For this analysis, I evaluated four machine learning models, each implemented with Python and the machine learning library `sklearn`.

- **Naïve (baseline) model** – predicts the mean `satisfaction` value for all observations.
- **Linear regression** – fits a linear relationship between the input features and the target variable by estimating a best-fit line (or hyperplane) through the data.
- **Random forest regression** – an ensemble learning method that combines many decision trees to improve predictive performance and reduce overfitting.
- **Recursive Feature Elimination (RFE)** – a random forest model combined with an iterative feature selection procedure that removes the least informative predictors to reduce the dimensionality of the input space.

Each model was trained on 80% of the data and evaluated on the remaining 20%. Model performance was evaluated primarily using Mean Squared Error (MSE).

3 Results

3.1 Descriptive Statistics

The distribution of the dependent variable is approximately normal with a slight left skew. The average `satisfaction` score was found at 5.96 on the 10 point scale.

Using Python and a combination of the `matplotlib` and `seaborn` libraries I produced many varieties of visualizations to understand the underlying variation within the data features. Some of the broader observations included fairly consistent sleep patterns and wearable device scores falling at around 80 on a scale to 100 for activity, sleep, and readiness.

3.2 Correlation Analysis

I ran a simple statistical correlation between `satisfaction` and each of the 28 columns in the dataset. These correlations are shown in Figure 7.

This analysis reveals, at best, relatively weak associations between the independent variables and the target variable. The strongest positive correlation was approximately 0.26, suggesting no single factor in isolation would be able to predict good days.

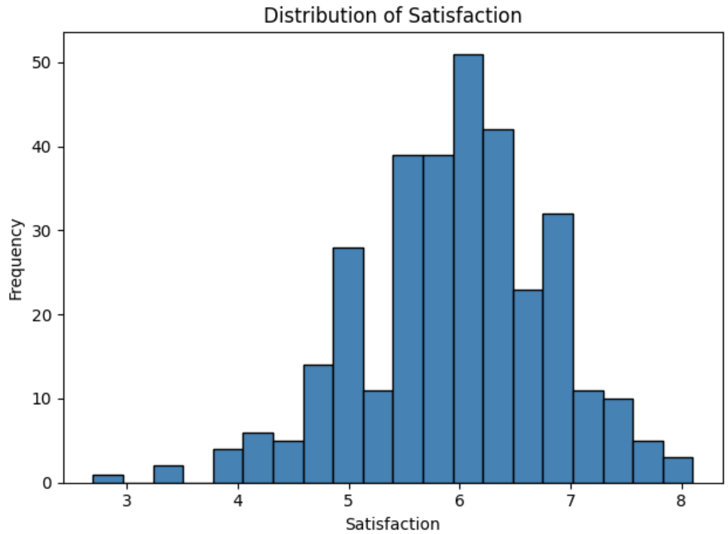

Fig. 6. Distribution of satisfaction scores, produced with `matplotlib`

Model	$\mathbf{R^2}$	**MSE**
Naïve baseline	-0.0253	0.9066
Linear regression	0.0869	0.8074
Random forest	0.1163	0.7814
Random forest with RFE	**0.1229**	**0.7756**

3.3 Model Performance

The best performing model was the model which used a random forest with recursive feature elimination (RFE). This reduced the input space to only the top 5 more predictive features and achieved a mean squared error of **0.776**.

The baseline model achieved a mean squared error of 0.906.

Comparing the RFE against baseline shows an improvement of 14.5%.

$$\frac{\mathrm{MSE}_{\mathrm{baseline}} - \mathrm{MSE}_{\mathrm{RFE}}}{\mathrm{MSE}_{\mathrm{baseline}}} = \frac{0.9066 - 0.7756}{0.9066} = 14.47\%$$

While this improvement demonstrates some level of predictability, the overall performance is modest, as shown in Figure 8.

3.4 Feature Importance

The recursive feature elimination identified the five variables that contributed the most to predictions of `satisfaction`. These are shown in Figure 9.

1. Active Calories

2. YouTube Consumption
3. Activity Score
4. Bedtime
5. Sleep Duration

Although the selected features were among the most highly correlated, they are not simply a list of the top five most correlated features. This implies that there is more meaningful separation between Active Calories (the most highly correlated) and the Activity Score than there is a distinction between Active Calories and Strength Workouts.

Using the `PartialDependencyDisplay` from the `sklearn` library, I plotted the function between domains of each of the five most predictive features and the predictive satisfaction across the range of each feature.

3.5 Behavioral Threshold Analysis

The partial dependency plots reveal some tentative behavioral patterns which were associated with meaningful shifts in `satisfaction`.

- Bedtime exhibits a "Goldilocks" optimal range, not too early, not too late
- Sleep duration is the most surprising, where less sleep yielded better satisfaction scores than more sleep
- Generally higher levels of activity, both in terms of Activity Score and Active Calories, is associated with higher satisfaction
- Finally, YouTube consumption exhibits a clear inverse relationship with satisfaction

4 Limitations

These findings and their interpretation are constrained by several factors.

First, these data represent the behavioral measures and observations of one single individual. This severely limits the generalizability of the results.

Second, despite covering five years, the dataset only contains a fairly modest sample size for machine learning analyses.

Third, the decision to publish these data and the resulting decision to sanitize through weekly averages has the effect of applying a smoothing function over the whole dataset. This greatly reduces the signal available for the machine learning model to pick up on. While this seemed to be a potential *benefit* up front, assuming that any results obtained after applying averages would be less likely to the variability of an individual day, but instead an aggregation of the results over the week, this had the effect of blunting the signal to the point where the naive model was actually *relatively strong*.

Finally, the subjective measures of satisfaction are noisy at best. The nightly grading of the days "goodness" is prone to drift and revert to the mean. Even if life is getting markedly better, the expectation of our lives grows with it, yielding a satisfaction score that more reflects the delta between daily expectations and daily results, as opposed to overall "life satisfaction".

5 Conclusion

This study examined whether the factors that reliably produce a "good day" can be determined through the use of longitudinal quantified self data describing an individual.

While the machine learning models were able to improve modestly upon a naive baseline model, the overall predictive power of the data remains limited. Only a small fraction of the variation in weekly average reported satisfaction was explainable with the inputs. Most of the features were not useful. The remaining useful features dealt with sleep, activity, and the engagement with passive media consumption.

Although the predictive power of this behavioral self-tracking was limited, this study demonstrates how quantified-self datasets can be used as a framework for carrying out empirical personal experimentation.

These results suggest that life satisfaction is not as simple as optimizing inputs. That which makes a satisfying life may not be easily tracked. Public speaker Cory House once remarked, *"The things most worth measuring are the hardest to measure."*

Future research in this area could benefit from a finer level of granularity, inputs from more than one individual, and larger datasets.

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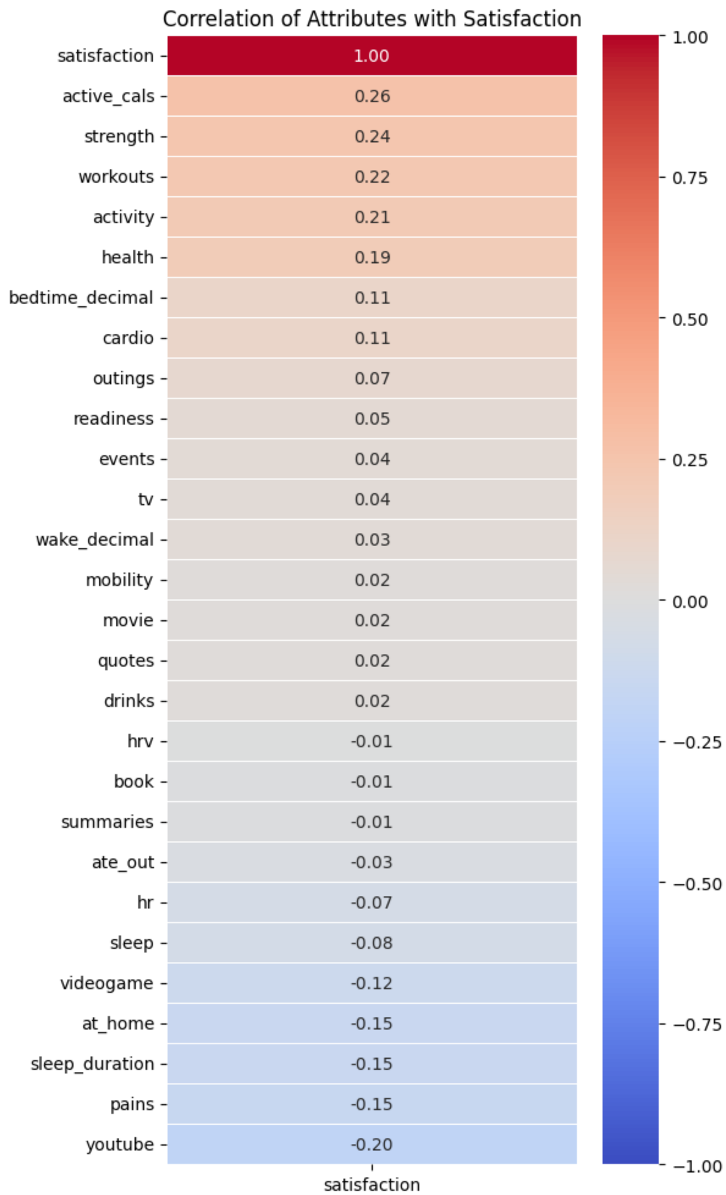

Fig. 7. Correlation matrix of variables, produced with `seaborn`

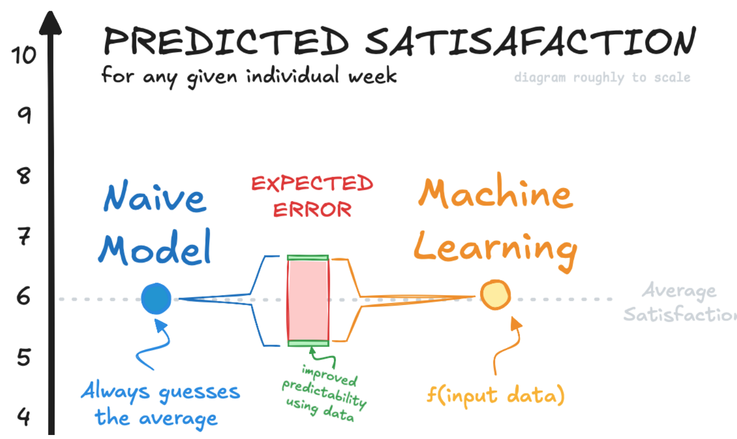

Fig. 8. Improvement in context, drawn with Excalidraw

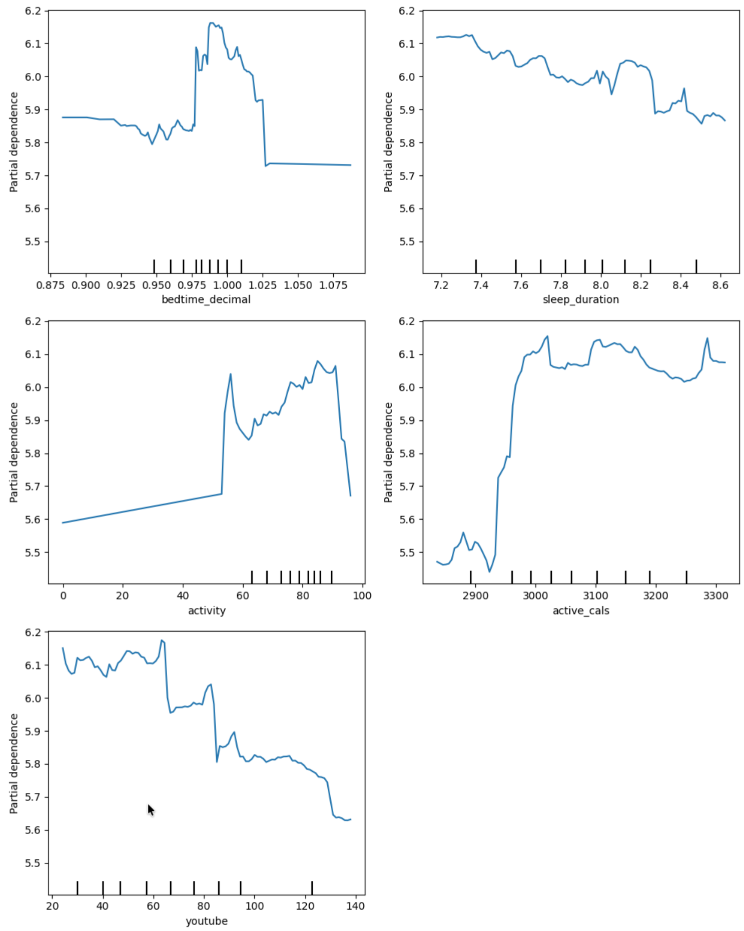

Fig. 9. Partial dependency plots, produced with `sklearn` and `matplotlib`